

LearnIT_homeworrk_005

▼ 1,2. W przeglądarce

httpbin.org ^{0.9.2}

[Base URL: httpbin.org/]

A simple HTTP Request & Response Service.

Run locally: `$ docker run -p 80:80 kennethreitz/httpbin`

[the developer - Website](#)

[Send email to the developer](#)

Schemes

HTTPS

HTTP Methods Testing different HTTP verbs

Auth Auth methods

Status codes Generates responses with given status code

Request inspection Inspect the request data

Response inspection Inspect the response data like caching and headers

Response formats Returns responses in different data formats

Dynamic data Generates random and dynamic data

DevTools is now available in Polish

[Don't show again](#) [Always match Chrome's language](#)

[Switch DevTools to Polish](#)

Elements Console Sources **Network** Performance >> 1

☐ Preserve log ☐ Disable cache No throttling

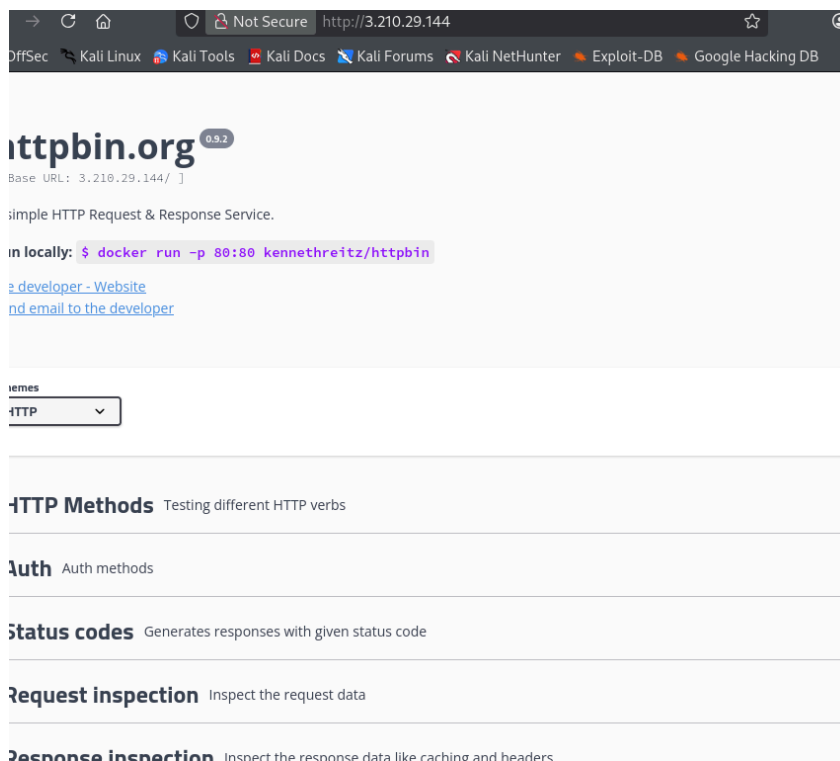
5,000 ms 10,000 ms 15,000 ms 20,000 ms 25,000 ms

Name	X	Headers	Preview	Response	Initiator	Time
▼ httpbin.org						
▼ General						
Request URL				https://httpbin.org/		
Request Method				GET		
Status Code				200 OK		
Remote Address				3.210.29.144:443		
Referrer Policy				strict-origin-when-cross-origi		
▼ Response headers						
Access-Control-Allow-Credentials				true		
Access-Control-Allow-Origin				*		
Content-Length				9593		
Content-Type				text/html; charset=utf-8		
Date				Thu, 20 Aug 2026 12:32:33 GMT		
Server				gunicorn/19.9.0		
▼ Request Headers						
:authority				httpbin.org		
:method				GET		
:path				/		
:scheme				https		
Accept				text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7		
Accept-Encoding				gzip, deflate, br, zstd		

request Get ze statusem 200 OK

Remote address (adres komputera, z którym host usiuguje się połączyć to: 3.210.29.144; :443

- ▼ wpisanie w pole adresu przeglądarki Remote address spowoduje nawiązanie połączenia z tą samą stroną)



▼ Response Headers

access-control-allow-credentials: **True** - response header informuje przeglądarkę o tym, czy serwer dopuszcza referencje użytkownika takie jak cookies, authorization headers lub certyfikat TLS.

access-control-allow-origin: * - precyzuje strony, które strony mają dostęp do danych strony. "*" - dopuszcza wszystkie strony

content length (rozmiar wiadomości w bajtach: 9593

content-type: text/html; charset=utf-8

date: Thu, 20 Aug 2026 12:32:33 GMT

server: gunicorn/19.9.0

connection (na Kali): keep-alive

▼ Request Headers

:**authority** - httpbin.org

:**method** - GET

:**path** - /

:**scheme** - https

accept -

text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signexchange;v=b3;q=0.7

accept-encoding - gzip, deflate, br, zstd - compression methodes the host supports

accept-language - pl-PL,pl;q=0.9,en-US;q=0.8,en;q=0.7

cache-control - max-age=0 - cache ulega przeterminowaniu po 0 sekundach, nie może być ponownie wykorzystany bez potwierdzenia ze strony serwera.

priority - u=0, i ; u - urgency, u=0 oznacza **najwyższy poziom pilności**, a obecność flagi "i" sygnalizuje, że zasób może być przetwarzany **inkrementalnie** (strumieniowo, zanim pobierze się w całości)

sec-ch-ua - "Not;A=Brand";v="8", "Chromium";v="150", "Google Chrome";v="150" - User-Agent Client Hint - informacja o marce i wersji przeglądarki internetowej

sec-ch-ua-mobile - ?0 - informuje, czy urządzenie, na którym zainstalowana jest przeglądarka, jest urządzeniem mobilnym

sec-ch-ua-platform - "Windows"

sec-fetch-dest - document

sec-fetch-mode - navigate

sec-fetch-site - same-origin

sec-fetch-user - ?1 - informuje, że akcja została zainicjowana przez fizycznego użytkownika

upgrade-insecure-requests - 1 - informuje, że host obsługuje i preferuje bezpieczne szyfrowane odpowiedzi HTTPS

user-agent - Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36

▼ 3. W terminalu - curl -v <https://httpbin.org/get>

```

Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. Wszelkie prawa zastrzeżone.

C:\Users\Lenovo>curl -v https://httpbin.org/get
* Host httpbin.org:443 was resolved.
* IPv6: (none)
* IPv4: 44.196.25.30
* Trying 44.196.25.30:443...
* schannel: disabled automatic use of client certificate
* ALPN: curl offers http/1.1
* ALPN: server accepted http/1.1
* Established connection to httpbin.org (44.196.25.30 port 443) from 192.168.0.136 port 59368
* using HTTP/1.x
> GET /get HTTP/1.1
> Host: httpbin.org
> User-Agent: curl/8.21.0
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Date: Thu, 20 Aug 2026 13:13:12 GMT
< Content-Type: application/json
< Content-Length: 253
< Connection: keep-alive
< Server: gunicorn/19.9.0
< Access-Control-Allow-Origin: *
< Access-Control-Allow-Credentials: true
<
{
  "args": {},
  "headers": {
    "Accept": "*/*",
    "Host": "httpbin.org",
    "User-Agent": "curl/8.21.0",
    "X-Amzn-Trace-Id": "Root=1-6a86fd68-38b1b80a47180d33452b8991"
  },
  "origin": "77.236.4.41",
  "url": "https://httpbin.org/get"
}
* Connection #0 to host httpbin.org:443 left intact

C:\Users\Lenovo>

```

origin: 44.196.25.30 (inny niż w przeglądarce, ale też łączy ze stroną

- Host httpbin.org:443 was resolved.
- IPv6: (none) - brak przypisanego adresu IPv6
- IPv4: 44.196.25.30
- Trying 44.196.25.30:443...
- schannel: disabled automatic use of client certificate - system jest skonfigurowany, aby nie wysyłać żądanych certyfikatów użytkownika automatycznie
- ALPN: curl offers http/1.1
- ALPN: server accepted http/1.1
- Established connection to httpbin.org (44.196.25.30 port 443) from 192.168.0.136 port 59368 -połączenie do portu 443 serwera z portu 59368 hosta
- using HTTP/1.x - używany jest protokół, który obsługuje jedno żądanie i jedną odpowiedź w danym momencie

```

GET /get HTTP/1.1
Host: httpbin.org
User-Agent: curl/8.21.0
Accept: /

```

- Request completely sent off
 - < HTTP/1.1 200 OK
 - < Date: Thu, 20 Aug 2026 13:13:12 GMT
 - < Content-Type: application/json
 - < Content-Length: 253 - rozmiar wiadomości
 - < Connection: keep-alive
 - < Server: gunicorn/19.9.0 - Gunicorn 'Green Unicorn' is a Python WSGI HTTP Server for UNIX.
 - < Access-Control-Allow-Origin: *
 - < Access-Control-Allow-Credentials: true
 - <
 - {
 - "args": {},
 - "headers": {
 - "Accept": "/",
 - "Host": "httpbin.org",
 - "User-Agent": "curl/8.21.0",
 - "X-Amzn-Trace-Id": "Root=1-6a86fd68-38b1b80a47180d33452b8991"
 - },
 - "origin": "77.236.4.41" - ?
 - "url": "<https://httpbin.org/get>"
 - }
- Connection #0 to host httpbin.org:443 left intact - powiązane z connection: keep-alive

▼ 4, 5 Wireshark - Learnit

1. Uruchom Wireshark z filtrem `udp.port == 53`.
2. W innym terminalu wykonaj `nslookup learnit.com.pl`

▼ Query: Flags: 0x0100 Standard query (IPv4)

1741	56.896978500	2a02:a302:0:1::10	2a02:a317:4adb:4b80...	DNS	212	Standard query response 0x0001 No such name PTR 0.1.0.0.0.0.0.0
1742	56.898783700	2a02:a317:4adb:4b80...	2a02:a302:0:1::10	DNS	99	Standard query 0x0002 A learnit.com.pl.home
1743	56.925851500	2a02:a302:0:1::10	2a02:a317:4adb:4b80...	DNS	174	Standard query response 0x0002 No such name A learnit.com.pl.home
1744	56.926519900	2a02:a317:4adb:4b80...	2a02:a302:0:1::10	DNS	99	Standard query 0x0003 AAAA learnit.com.pl.home
1747	56.952651400	2a02:a302:0:1::10	2a02:a317:4adb:4b80...	DNS	174	Standard query response 0x0003 No such name AAAA learnit.com.pl.h
1748	56.953597100	2a02:a317:4adb:4b80...	2a02:a302:0:1::10	DNS	94	Standard query 0x0004 A learnit.com.pl
1749	57.029122900	2a02:a302:0:1::10	2a02:a317:4adb:4b80...	DNS	110	Standard query response 0x0004 A learnit.com.pl A 142.93.161.101
1750	57.033544700	2a02:a317:4adb:4b80...	2a02:a302:0:1::10	DNS	94	Standard query 0x0005 AAAA learnit.com.pl
1754	57.111087000	2a02:a302:0:1::10	2a02:a317:4adb:4b80...	DNS	165	Standard query response 0x0005 AAAA learnit.com.pl SOA ns75.domain

> Frame 1742: Packet, 99 bytes on wire (792 bits), 99 bytes captured (792 bits) on interface \Device\NPF...		0000	54 67 51 2b 76 66 f0 20 ff 3a 35 b3 86 c
> Ethernet II, Src: Intel_3a:35:b3 (f0:20:ff:3a:35:b3), Dst: CompalBroadb_2b:76:66 (54:67:51:2b:76:66)		0010	ae ea 00 2d 11 40 2a 02 a3 17 4a db 4b 80
> Internet Protocol Version 6, Src: 2a02:a317:4adb:4b80:1006:9539:328c:56d5, Dst: 2a02:a302:0:1::10		0020	95 39 32 8c 56 d5 2a 02 a3 02 00 00 00 00
> User Datagram Protocol, Src Port: 53667, Dst Port: 53		0030	00 00 00 00 00 10 d1 a3 00 35 00 2d 5f c
▼ Domain Name System (query)		0040	01 00 00 01 00 00 00 00 00 00 07 6c 65 c
Transaction ID: 0x0002		0050	69 74 03 63 6f 6d 02 70 6c 04 68 6f 6d c
▼ Flags: 0x0100 Standard query		0060	01 00 01
0... .. = Response: Message is a query			
.000 0... .. = Opcode: Standard query (0)			
... ..0. = Truncated: Message is not truncated			
... ..1. = Recursion desired: Do query recursively			
... ..0. = Z: reserved (0)			
... ..0. = Non-authenticated data: Unacceptable			
Questions: 1			
Answer RRs: 0			
Authority RRs: 0			
Additional RRs: 0			
▼ Queries			
▼ learnit.com.pl.home: type A, class IN			
Name: learnit.com.pl.home			
[Name Length: 19]			
[Label Count: 4]			

▼ Response: Flags: 0x8183 Standard query response, No such name

1742	56.898783700	2a02:a317:4adb:4b80...	2a02:a302:0:1::10	DNS	99	Standard query 0x0002 A learnit.com.pl.home
1743	56.925851500	2a02:a302:0:1::10	2a02:a317:4adb:4b80...	DNS	174	Standard query response 0x0002 No such name A learnit.com.pl.home SOA a.root-servers.net
1744	56.926519900	2a02:a317:4adb:4b80...	2a02:a302:0:1::10	DNS	99	Standard query 0x0003 AAAA learnit.com.pl.home
1747	56.952651400	2a02:a302:0:1::10	2a02:a317:4adb:4b80...	DNS	174	Standard query response 0x0003 No such name AAAA learnit.com.pl.home SOA a.root-servers.net
1748	56.953597100	2a02:a317:4adb:4b80...	2a02:a302:0:1::10	DNS	94	Standard query 0x0004 A learnit.com.pl
1749	57.029122900	2a02:a302:0:1::10	2a02:a317:4adb:4b80...	DNS	110	Standard query response 0x0004 A learnit.com.pl A 142.93.161.101
1750	57.033544700	2a02:a317:4adb:4b80...	2a02:a302:0:1::10	DNS	94	Standard query 0x0005 AAAA learnit.com.pl
1754	57.111087000	2a02:a302:0:1::10	2a02:a317:4adb:4b80...	DNS	165	Standard query response 0x0005 AAAA learnit.com.pl SOA ns75.domaincontrol.com

> Frame 1743: Packet, 174 bytes on wire (1392 bits), 174 bytes captured (1392 bits) on interface \Device\NPF...		0000	f0 20 ff 3a 35 b3 54 67 51 2b 76 66 86 dd 60 03
> Ethernet II, Src: CompalBroadb_2b:76:66 (54:67:51:2b:76:66), Dst: Intel_3a:35:b3 (f0:20:ff:3a:35:b3)		0010	8e 81 00 78 11 39 2a 02 a3 02 00 00 00 01 00 00
> Internet Protocol Version 6, Src: 2a02:a302:0:1::10, Dst: 2a02:a317:4adb:4b80:1006:9539:328c:56d5		0020	00 00 00 00 00 10 2a 02 a3 17 4a db 4b 80 10 06
> User Datagram Protocol, Src Port: 53, Dst Port: 53667		0030	95 39 32 8c 56 d5 00 35 d1 a3 00 70 85 76 00 02
▼ Domain Name System (response)		0040	81 83 00 01 00 00 00 01 00 00 07 6c 65 61 72 6e
Transaction ID: 0x0002		0050	69 74 03 63 6f 6d 02 70 6c 04 68 6f 6d 65 00 00
▼ Flags: 0x8183 Standard query response, No such name		0060	01 00 01 00 00 06 00 01 00 01 51 7a 00 40 01 61
1... .. = Response: Message is a response		0070	0c 72 6f 6f 74 2d 73 65 72 76 65 72 73 03 6e 65
.000 0... .. = Opcode: Standard query (0)		0080	74 00 05 06 73 74 6c 64 6c 76 65 72 69 73 69 67
... ..0. = Authoritative: Server is not an authority for domain		0090	6e 2d 67 72 73 03 63 6f 6d 00 78 c3 8e 00 00 00
... ..1. = Recursion desired: Do query recursively		00a0	07 08 00 00 03 84 00 09 3a 80 00 01 51 80
... ..1... .. = Recursion available: Server can do recursive queries			
... ..0. = Z: reserved (0)			
... ..0. = Answer authenticated: Answer/authority portion was not authenticated by th			
... ..0. = Non-authenticated data: Unacceptable			
... ..0011 = Reply code: No such name (3)			
Questions: 1			
Answer RRs: 0			
Authority RRs: 1			
Additional RRs: 0			
▼ Queries			

▼ Query: Flags: 0x0100 Standard query (IPv6)

1742 56.898783700	2a02:a317:4adb:4b80::	2a02:a302:0:1:1:10	DNS	99 Standard query 0x0002 A learnit.com.pl.home
1743 56.925851500	2a02:a302:0:1:1:10	2a02:a317:4adb:4b80::	DNS	174 Standard query response 0x0002 No such name A learnit.com.pl.home SOA a.root-servers.net
1744 56.926519900	2a02:a317:4adb:4b80::	2a02:a302:0:1:1:10	DNS	99 Standard query 0x0003 AAAA learnit.com.pl.home
1747 56.952651400	2a02:a302:0:1:1:10	2a02:a317:4adb:4b80::	DNS	174 Standard query response 0x0003 No such name AAAA learnit.com.pl.home SOA a.root-servers.net
1748 56.953597100	2a02:a317:4adb:4b80::	2a02:a302:0:1:1:10	DNS	94 Standard query 0x0004 A learnit.com.pl
1749 57.029122900	2a02:a302:0:1:1:10	2a02:a317:4adb:4b80::	DNS	110 Standard query response 0x0004 A learnit.com.pl A 142.93.161.101
1750 57.033544700	2a02:a317:4adb:4b80::	2a02:a302:0:1:1:10	DNS	94 Standard query 0x0005 AAAA learnit.com.pl
1754 57.111087000	2a02:a302:0:1:1:10	2a02:a317:4adb:4b80::	DNS	165 Standard query response 0x0005 AAAA learnit.com.pl SOA ns75.domaincontrol.com

Frame 1744: Packet, 99 bytes on wire (792 bits), 99 bytes captured (792 bits) on interface \Device\NPF...	0000 54 67 51 2b 76 66 f0 20 ff 3a 35 b3 86 dd 60 03 TgQ+vf...:15...
Ethernet II, Src: Intel_3a:35:b3 (f0:20:ff:3a:35:b3), Dst: CompalBroadb_2b:76:66 (54:67:51:2b:76:66)	0010 67 cf 00 2d 11 40 2a 02 a3 17 4a db 4b 80 10 06 g...@*...:J:K...
Internet Protocol Version 6, Src: 2a02:a317:4adb:4b80:1006:9539:328c:56d5, Dst: 2a02:a302:0:1:1:10	0020 95 39 32 8c 56 d5 2a 02 a3 02 00 00 00 01 00 00 -92-V:*...:.....
User Datagram Protocol, Src Port: 53668, Dst Port: 53	0030 00 00 00 00 00 10 d1 a4 00 35 00 2d 5f 6a 00 035...j...
Domain Name System (query)	0040 01 00 00 01 00 00 00 00 00 07 6c 65 61 72 6elear...
Transaction ID: 0x0003	0050 69 74 03 63 6f 6d 02 70 6c 04 68 6f 6d 65 00 00 it-com p l-home...
Flags: 0x0100 Standard query	0060 1c 00 01 ...
0... .. = Response: Message is a query	
.000 0... .. = Opcode: Standard query (0)	
... ..0... .. = Truncated: Message is not truncated	
... ..1... .. = Recursion desired: Do query recursively	
... ..0... .. = Z: reserved (0)	
... ..0... .. = Non-authenticated data: Unacceptable	
Questions: 1	
Answer RRs: 0	
Authority RRs: 0	
Additional RRs: 0	
Queries	
learnit.com.pl.home: type AAAA, class IN	
Name: learnit.com.pl.home	
[Name Length: 19]	
[Label Count: 4]	

▼ Response: Flags: 0x8183 Standard query response, No such name

1742 56.898783700	2a02:a317:4adb:4b80::	2a02:a302:0:1:1:10	DNS	99 Standard query 0x0002 A learnit.com.pl.home
1743 56.925851500	2a02:a302:0:1:1:10	2a02:a317:4adb:4b80::	DNS	174 Standard query response 0x0002 No such name A learnit.com.pl.home SOA a.root-servers.net
1744 56.926519900	2a02:a317:4adb:4b80::	2a02:a302:0:1:1:10	DNS	99 Standard query 0x0003 AAAA learnit.com.pl.home
1747 56.952651400	2a02:a302:0:1:1:10	2a02:a317:4adb:4b80::	DNS	174 Standard query response 0x0003 No such name AAAA learnit.com.pl.home SOA a.root-servers.net
1748 56.953597100	2a02:a317:4adb:4b80::	2a02:a302:0:1:1:10	DNS	94 Standard query 0x0004 A learnit.com.pl
1749 57.029122900	2a02:a302:0:1:1:10	2a02:a317:4adb:4b80::	DNS	110 Standard query response 0x0004 A learnit.com.pl A 142.93.161.101
1750 57.033544700	2a02:a317:4adb:4b80::	2a02:a302:0:1:1:10	DNS	94 Standard query 0x0005 AAAA learnit.com.pl
1754 57.111087000	2a02:a302:0:1:1:10	2a02:a317:4adb:4b80::	DNS	165 Standard query response 0x0005 AAAA learnit.com.pl SOA ns75.domaincontrol.com

Frame 1747: Packet, 174 bytes on wire (1392 bits), 174 bytes captured (1392 bits) on interface \Device\NPF...	0000 70 20 ff 3a 35 b3 54 67 51 2b 76 66 50 dd 60 04 :S-Tg Qevf...
Ethernet II, Src: CompalBroadb_2b:76:66 (54:67:51:2b:76:66), Dst: Intel_3a:35:b3 (f0:20:ff:3a:35:b3)	0010 7c 05 00 78 11 39 2a 02 a3 02 00 00 00 01 00 00 --x-S*...:.....
Internet Protocol Version 6, Src: 2a02:a302:0:1:1:10, Dst: 2a02:a317:4adb:4b80:1006:9539:328c:56d5	0020 00 00 00 00 10 2a 02 a3 17 4a db 4b 80 10 06:J:K...
User Datagram Protocol, Src Port: 53, Dst Port: 53668	0030 95 39 32 8c 56 d5 00 35 d1 a4 00 78 6a 96 00 03 -92-V-*...:.....
Domain Name System (response)	0040 01 00 00 00 00 00 01 00 00 07 6c 65 61 72 6elear...
Transaction ID: 0x0003	0050 69 74 03 63 6f 6d 02 70 6c 04 68 6f 6d 65 00 00 it-com p l-home...
Flags: 0x8183 Standard query response, No such name	0060 0c 72 6f 6f 74 2d 73 65 72 76 65 72 73 6a 65 root-se rvers ne...
Questions: 1	0070 00 00 00 27 74 6c 64 76 65 72 69 73 69 6f n-mla3 versig...
Answer RRs: 1	0080 6a 2d 67 72 73 03 63 6f 6d 00 78 6c 8a 00 00 -grn co m x:15g...
Authority RRs: 0	0090 07 00 00 00 05 04 00 00 5a 00 00 01 91 00:~:~-q...
Additional RRs: 0	
Queries	
learnit.com.pl.home: type AAAA, class IN	
Name: learnit.com.pl.home	
[Name Length: 19]	
[Label Count: 4]	
Type: AAAA (28) (IPv6 Address)	
Class: IN (0x0001)	
Authoritative nameservers	
[Request ID: 1748]	
[Time: 26.139500 milliseconds]	

▼ Query: Flags: 0x0100 Standard query (IPv4)

1742 56.898783700	2a02:a317:4adb:4b80::	2a02:a302:0:1:1:10	DNS	99 Standard query 0x0002 A learnit.com.pl.home
1743 56.925851500	2a02:a302:0:1:1:10	2a02:a317:4adb:4b80::	DNS	174 Standard query response 0x0002 No such name A learnit.com.pl.home SOA a.root-servers.net
1744 56.926519900	2a02:a317:4adb:4b80::	2a02:a302:0:1:1:10	DNS	99 Standard query 0x0003 AAAA learnit.com.pl.home
1747 56.952651400	2a02:a302:0:1:1:10	2a02:a317:4adb:4b80::	DNS	174 Standard query response 0x0003 No such name AAAA learnit.com.pl.home SOA a.root-servers.net
1748 56.953597100	2a02:a317:4adb:4b80::	2a02:a302:0:1:1:10	DNS	94 Standard query 0x0004 A learnit.com.pl
1749 57.029122900	2a02:a302:0:1:1:10	2a02:a317:4adb:4b80::	DNS	110 Standard query response 0x0004 A learnit.com.pl A 142.93.161.101
1750 57.033544700	2a02:a317:4adb:4b80::	2a02:a302:0:1:1:10	DNS	94 Standard query 0x0005 AAAA learnit.com.pl
1754 57.111087000	2a02:a302:0:1:1:10	2a02:a317:4adb:4b80::	DNS	165 Standard query response 0x0005 AAAA learnit.com.pl SOA ns75.domaincontrol.com

Frame 1748: Packet, 94 bytes on wire (752 bits), 94 bytes captured (752 bits) on interface \Device\NPF...	0000 54 67 51 2b 76 66 f0 20 ff 3a 35 b3 86 dd 60 03 TgQ+vf...:15...
Ethernet II, Src: Intel_3a:35:b3 (f0:20:ff:3a:35:b3), Dst: CompalBroadb_2b:76:66 (54:67:51:2b:76:66)	0010 70 90 20 11 40 2a 02 a3 17 4a db 4b 80 10 06 pTcfp...:74...
Internet Protocol Version 6, Src: 2a02:a317:4adb:4b80:1006:9539:328c:56d5, Dst: 2a02:a302:0:1:1:10	0020 95 39 32 8c 56 d5 2a 02 a3 02 00 00 00 01 00 00 -92-V-*...:.....
User Datagram Protocol, Src Port: 53669, Dst Port: 53	0030 00 00 00 00 00 10 d1 a4 00 35 00 2d 5f 6a 00 035...j...
Domain Name System (query)	0040 01 00 00 01 00 00 00 00 00 07 6c 65 61 72 6elear...
Transaction ID: 0x0004	0050 69 74 03 63 6f 6d 02 70 6c 04 68 6f 6d 65 00 00 it-com p l-home...
Flags: 0x0100 Standard query	0060 1c 00 01 ...
0... .. = Response: Message is a query	
.000 0... .. = Opcode: Standard query (0)	
... ..0... .. = Truncated: Message is not truncated	
... ..1... .. = Recursion desired: Do query recursively	
... ..0... .. = Z: reserved (0)	
... ..0... .. = Non-authenticated data: Unacceptable	
Questions: 1	
Answer RRs: 0	
Authority RRs: 0	
Additional RRs: 0	
Queries	
learnit.com.pl: type A, class IN	
Name: learnit.com.pl	
[Name Length: 14]	
[Label Count: 3]	
Type: A (1) (IPv4 Address)	
Class: IN (0x0001)	
[Request ID: 1749]	

▼ Flags: 0x8180 Standard query response, No error

1742 56.898783700	2a02:a317:4adb:4b80::	2a02:a302:0:1::10	DNS	99 Standard query 0x0002 A learnit.com.pl.home
1743 56.925851500	2a02:a302:0:1::10	2a02:a317:4adb:4b80::	DNS	174 Standard query response 0x0002 No such name A learnit.com.pl.home SOA a.root-servers.net
1744 56.926519900	2a02:a317:4adb:4b80::	2a02:a302:0:1::10	DNS	99 Standard query 0x0003 AAAA learnit.com.pl.home
1747 56.952651400	2a02:a302:0:1::10	2a02:a317:4adb:4b80::	DNS	99 Standard query response 0x0003 No such name AAAA learnit.com.pl.home SOA a.root-servers.net
1748 56.953597100	2a02:a317:4adb:4b80::	2a02:a302:0:1::10	DNS	94 Standard query 0x0004 A learnit.com.pl
1749 57.029122900	2a02:a302:0:1::10	2a02:a317:4adb:4b80::	DNS	110 Standard query response 0x0004 A learnit.com.pl A 142.93.161.101
1750 57.033544700	2a02:a317:4adb:4b80::	2a02:a302:0:1::10	DNS	94 Standard query 0x0005 AAAA learnit.com.pl
1754 57.111807000	2a02:a302:0:1::10	2a02:a317:4adb:4b80::	DNS	165 Standard query response 0x0005 AAAA learnit.com.pl SOA ns75.domaincontrol.com

> Frame 1749: Packet, 110 bytes on wire (880 bits), 110 bytes captured (880 bits) on interface \Device\NPF{...} > Ethernet II, Src: CompalBroadb_2b:76:66 (54:67:51:2b:76:66), Dst: Intel_3a:35:b3 (f0:20:ff:3a:35:b3) > Internet Protocol Version 6, Src: 2a02:a302:0:1::10, Dst: 2a02:a317:4adb:4b80:1006:9539:328c:56d5 > User Datagram Protocol, Src Port: 53, Dst Port: 53669 ▼ Domain Name System (response) Transaction ID: 0x0004 Flags: 0x8180 Standard query response, No error 1... .. = Response: Message is a response .000 0... .. = Opcode: Standard query (0) 0... .. = Authoritative: Server is not an authority for domain 0... .. = Truncated: Message is not truncated 1... .. = Recursion desired: Do query recursively 1... .. = Recursion available: Server can do recursive queries 0... .. = Z: reserved (0) 0... .. = Answer authenticated: Answer/authority portion was not authenticated by the 0... .. = Non-authenticated data: Unacceptable 0000 = Reply code: No error (0) Questions: 1 Answer RRs: 1 Authority RRs: 0 Additional RRs: 0 ▼ Queries ▼ learnit.com.pl: type A, class IN Name: learnit.com.pl [Name Length: 14] [Label Count: 3] Type: A (1) (Host Address) Class: IN (0x0001) ▼ Answers [Request In: 1748] [Time: 75.525800 milliseconds]	0000 f0 20 ff 3a 35 b3 54 67 51 2b 76 66 dd 60 09 ...:5 Tg Qrvf... 0010 62 30 00 38 11 39 2a 02 a3 02 00 00 01 00 00 b0 8 9... 0020 00 00 00 00 00 10 2a 02 a3 17 4a db 4b 80 10 06 ...:K... 0030 95 39 32 8c 56 d5 00 35 d1 a5 00 38 30 d0 00 04 ...2 V 5 ...80... 0040 81 80 00 01 00 01 00 00 00 00 07 6c 65 61 72 6e ...learn 0050 69 74 03 63 6f 6d 02 70 6c 00 00 01 00 01 c0 0c it.com p 1... 0060 00 01 00 01 00 00 02 58 00 04 8e 5d a1 65 ...X ...]e
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▼ Query: Flags: 0x0100 Standard query (IPv6)

1742 56.898783700	2a02:a317:4adb:4b80::	2a02:a302:0:1::10	DNS	99 Standard query 0x0002 A learnit.com.pl.home
1743 56.925851500	2a02:a302:0:1::10	2a02:a317:4adb:4b80::	DNS	174 Standard query response 0x0002 No such name A learnit.com.pl.home SOA a.root-servers.net
1744 56.926519900	2a02:a317:4adb:4b80::	2a02:a302:0:1::10	DNS	99 Standard query 0x0003 AAAA learnit.com.pl.home
1747 56.952651400	2a02:a302:0:1::10	2a02:a317:4adb:4b80::	DNS	174 Standard query response 0x0003 No such name AAAA learnit.com.pl.home SOA a.root-servers.net
1748 56.953597100	2a02:a317:4adb:4b80::	2a02:a302:0:1::10	DNS	94 Standard query 0x0004 A learnit.com.pl
1749 57.029122900	2a02:a302:0:1::10	2a02:a317:4adb:4b80::	DNS	110 Standard query response 0x0004 A learnit.com.pl A 142.93.161.101
1750 57.033544700	2a02:a317:4adb:4b80::	2a02:a302:0:1::10	DNS	94 Standard query 0x0005 AAAA learnit.com.pl
1754 57.111807000	2a02:a302:0:1::10	2a02:a317:4adb:4b80::	DNS	165 Standard query response 0x0005 AAAA learnit.com.pl SOA ns75.domaincontrol.com

> Frame 1750: Packet, 94 bytes on wire (752 bits), 94 bytes captured (752 bits) on interface \Device\NPF{...} > Ethernet II, Src: Intel_3a:35:b3 (f0:20:ff:3a:35:b3), Dst: CompalBroadb_2b:76:66 (54:67:51:2b:76:66) > Internet Protocol Version 6, Src: 2a02:a317:4adb:4b80:1006:9539:328c:56d5, Dst: 2a02:a302:0:1::10 > User Datagram Protocol, Src Port: 53670, Dst Port: 53 ▼ Domain Name System (query) Transaction ID: 0x0005 Flags: 0x0100 Standard query 0... .. = Response: Message is a query .000 0... .. = Opcode: Standard query (0) 0... .. = Truncated: Message is not truncated 1... .. = Recursion desired: Do query recursively 0... .. = Z: reserved (0) 0... .. = Non-authenticated data: Unacceptable Questions: 1 Answer RRs: 0 Authority RRs: 0 Additional RRs: 0 ▼ Queries ▼ learnit.com.pl: type AAAA, class IN Name: learnit.com.pl [Name Length: 14] [Label Count: 3] Type: AAAA (28) (IP6 Address) Class: IN (0x0001) [Response In: 1754]	0000 54 67 51 2b 76 66 f0 20 ff 3a 35 b3 86 dd 60 02 TgQ+vf: ...:5... 0010 cf c9 00 28 11 40 2a 02 a3 17 4a db 4b 80 10 06 ...: (@* ...:J K... 0020 95 39 32 8c 56 d5 2a 02 a3 02 00 00 01 00 00 92 V+... 0030 00 00 00 00 00 10 d1 a6 00 35 00 28 5f 65 00 05 ...:5 (_... 0040 01 00 00 01 00 00 00 00 00 00 07 6c 65 61 72 6e ...learn 0050 69 74 03 63 6f 6d 02 70 6c 00 00 1c 00 01 it.com p 1...
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▼ Response: Flags: 0x8180 Standard query response, No error

1743 56.925851500 2a02:a302:0:1::10 2a02:a317:4adb:4b80:: DNS 174 Standard query response 0x0002 No such name A learnit.com.pl.home SOA a.root-servers.net
1744 56.926519900 2a02:a317:4adb:4b80:: 2a02:a302:0:1::10 DNS 99 Standard query 0x0003 AAAA learnit.com.pl.home
1747 56.952651400 2a02:a302:0:1::10 2a02:a317:4adb:4b80:: DNS 174 Standard query response 0x0003 No such name AAAA learnit.com.pl.home SOA a.root-servers.net
1748 56.953597100 2a02:a317:4adb:4b80:: 2a02:a302:0:1::10 DNS 94 Standard query 0x0004 A learnit.com.pl
1749 57.029122900 2a02:a302:0:1::10 2a02:a317:4adb:4b80:: DNS 110 Standard query response 0x0004 A learnit.com.pl A 142.93.161.101
1750 57.033544700 2a02:a317:4adb:4b80:: 2a02:a302:0:1::10 DNS 94 Standard query 0x0005 AAAA learnit.com.pl
1754 57.111087000 2a02:a302:0:1::10 2a02:a317:4adb:4b80:: DNS 165 Standard query response 0x0005 AAAA learnit.com.pl SOA ns75.domaincontrol.com

> Frame 1754: Packet, 165 bytes on wire (1320 bits), 165 bytes captured (1320 bits) on interface DeviceN
> Ethernet II, Src: CompalBroadb_2b:76:66 (54:67:51:2b:76:66), Dst: Intel_3a:35:b3 (f0:20:ff:3a:35:b3)
> Internet Protocol Version 6, Src: 2a02:a302:0:1::10, Dst: 2a02:a317:4adb:4b80:1006:9539:328c:56d5
> User Datagram Protocol, Src Port: 53, Dst Port: 53670
▼ Domain Name System (response)
Transaction ID: 0x0005
▼ Flags: 0x8180 Standard query response, No error
1... .. = Response: Message is a response
.000 0... .. = Opcode: Standard query (0)
.... 0... .. = Authoritative: Server is not an authority for domain
.... 0... .. = Truncated: Message is not truncated
.... 1... .. = Recursion desired: Do query recursively
.... 1... .. = Recursion available: Server can do recursive queries
.... 0... .. = Z: reserved (0)
.... 0... .. = Answer authenticated: Answer/authority portion was not authenticated by the
.... 0... .. = Non-authenticated data: Unacceptable
.... 0000 = Reply code: No error (0)
Questions: 1
Answer RRs: 0
Authority RRs: 1
Additional RRs: 0
▼ Queries
▼ learnit.com.pl: type AAAA, class IN
Name: learnit.com.pl
[Name Length: 14]
[Label Count: 3]
Type: AAAA (28) (IP6 Address)
Class: IN (0x0001)
> Authoritative nameservers
[Request In: 1750]
[Time: 77.542300 milliseconds]

0000 f0 20 ff 3a 35 b3 54 67 51 2b 76 66 86 dd 60 04 ...:5 Tg Q+vf...
0010 aa c8 00 6f 11 39 2a 02 a3 02 00 00 00 01 00 00 ...o 9*
0020 00 00 00 00 00 10 2a 02 a3 17 4a db 4b 80 10 06*...J K...
0030 95 39 32 8c 56 d5 00 35 d1 a6 00 6f c4 ae 00 05 ...92 V...5 ...o...
0040 00 00 00 00 00 00 01 00 00 07 6c 65 61 72 6e ...learn
0050 69 74 03 63 6f 6d 02 70 6c 00 00 1c 00 01 c0 0c ...com-p l.....
0060 00 06 00 01 00 00 02 58 00 3b 04 6e 73 37 35 0dX...ns75...
0070 64 6f 6d 61 69 6e 63 6f 6e 74 72 6f 6c 03 63 6f domainco ntrol.co
0080 6d 00 03 64 6e 73 05 6a 6f 6d 61 78 03 6e 65 74 m--dns-j omox-net
0090 00 78 c3 19 3c 00 00 70 80 00 00 1c 20 00 09 3a ...x<<<-p:t
00a0 80 00 00 02 58X

▼ 6. dns.qry.name contains "learnit"

Much easier to find.

No.	Time	Source	Destination	Protocol	Length	Info
1742	56.896703700	2a02:a317:4adb:4b80::	2a02:a302:0:1::10	DNS	99	Standard query 0x0002 A learnit.com.pl.home
1743	56.925851500	2a02:a302:0:1::10	2a02:a317:4adb:4b80::	DNS	174	Standard query response 0x0002 No such name A learnit.com.pl.home SOA a.root-servers.net
1744	56.926519900	2a02:a317:4adb:4b80::	2a02:a302:0:1::10	DNS	99	Standard query 0x0003 AAAA learnit.com.pl.home
1747	56.952651400	2a02:a302:0:1::10	2a02:a317:4adb:4b80::	DNS	174	Standard query response 0x0003 No such name AAAA learnit.com.pl.home SOA a.root-servers.net
1748	56.953597100	2a02:a317:4adb:4b80::	2a02:a302:0:1::10	DNS	94	Standard query 0x0004 A learnit.com.pl
1749	57.029122900	2a02:a302:0:1::10	2a02:a317:4adb:4b80::	DNS	110	Standard query response 0x0004 A learnit.com.pl A 142.93.161.101
1750	57.033544700	2a02:a317:4adb:4b80::	2a02:a302:0:1::10	DNS	94	Standard query 0x0005 AAAA learnit.com.pl
1754	57.111087000	2a02:a302:0:1::10	2a02:a317:4adb:4b80::	DNS	165	Standard query response 0x0005 AAAA learnit.com.pl SOA ns75.domaincontrol.com

▼ 7. Przeanalizuj three-way handshake filtrem tcp.flags.syn == 1.

nie widać TWH ponieważ ostatnia flaga [ACK] jest niewidoczna.

tcp.flags.syn == 1						
No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	2a02:a317:4adb:4b80::	2a02:a302:0:1::10	TCP	86	51382 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
2	0.000183500	2a02:a317:4adb:4b80::	2a02:a302:0:1::10	TCP	86	64967 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
3	0.000277100	2a02:a317:4adb:4b80::	2a02:a302:0:1::10	TCP	86	63721 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
4	0.025185200	2a02:a302:0:1::10	2a02:a317:4adb:4b80::	TCP	86	53 → 51382 [SYN, ACK] Seq=0 Ack=1 Win=64800 Len=0 MSS=1440 SACK_PERM WS=128
5	0.025185800	2a02:a302:0:1::10	2a02:a317:4adb:4b80::	TCP	86	53 → 63721 [SYN, ACK] Seq=0 Ack=1 Win=64800 Len=0 MSS=1440 SACK_PERM WS=128
12	0.026549900	2a02:a302:0:1::10	2a02:a317:4adb:4b80::	TCP	86	53 → 64967 [SYN, ACK] Seq=0 Ack=1 Win=64800 Len=0 MSS=1440 SACK_PERM WS=128
156	3.973705700	2a02:a317:4adb:4b80::	2a02:a302:0:1::10	TCP	86	55137 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
157	3.973842700	2a02:a317:4adb:4b80::	2a02:a302:0:1::10	TCP	86	64704 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
158	3.973920400	2a02:a317:4adb:4b80::	2a02:a302:0:1::10	TCP	86	58924 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
159	3.998833000	2a02:a302:0:1::10	2a02:a317:4adb:4b80::	TCP	86	53 → 64704 [SYN, ACK] Seq=0 Ack=1 Win=64800 Len=0 MSS=1440 SACK_PERM WS=128
160	3.998833400	2a02:a302:0:1::10	2a02:a317:4adb:4b80::	TCP	86	53 → 55137 [SYN, ACK] Seq=0 Ack=1 Win=64800 Len=0 MSS=1440 SACK_PERM WS=128
161	3.998833600	2a02:a302:0:1::10	2a02:a317:4adb:4b80::	TCP	86	53 → 58924 [SYN, ACK] Seq=0 Ack=1 Win=64800 Len=0 MSS=1440 SACK_PERM WS=128
953	40.173799900	192.168.0.136	52.168.117.174	TCP	66	50156 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
954	40.293244500	52.168.117.174	192.168.0.136	TCP	66	443 → 50156 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1420 WS=256 SACK_PERM
1340	57.090869100	2a02:a317:4adb:4b80::	2a02:a302:0:1::10	TCP	86	60241 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
1341	57.091037000	2a02:a317:4adb:4b80::	2a02:a302:0:1::10	TCP	86	54298 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
1342	57.091112000	2a02:a317:4adb:4b80::	2a02:a302:0:1::10	TCP	86	55133 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
1343	57.112296800	2a02:a302:0:1::10	2a02:a317:4adb:4b80::	TCP	86	53 → 60241 [SYN, ACK] Seq=0 Ack=1 Win=64800 Len=0 MSS=1440 SACK_PERM WS=128
1347	57.118773600	2a02:a302:0:1::10	2a02:a317:4adb:4b80::	TCP	86	53 → 54298 [SYN, ACK] Seq=0 Ack=1 Win=64800 Len=0 MSS=1440 SACK_PERM WS=128
1348	57.118774700	2a02:a302:0:1::10	2a02:a317:4adb:4b80::	TCP	86	53 → 55133 [SYN, ACK] Seq=0 Ack=1 Win=64800 Len=0 MSS=1440 SACK_PERM WS=128
1381	57.215763300	192.168.0.136	34.110.164.207	TCP	66	56894 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
1391	57.259202300	34.110.164.207	192.168.0.136	TCP	66	443 → 56894 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM WS=256
1525	63.989033300	2a02:a317:4adb:4b80::	2a02:a302:0:1::10	TCP	86	58919 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
1526	63.989191900	2a02:a317:4adb:4b80::	2a02:a302:0:1::10	TCP	86	61180 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
1527	63.989268600	2a02:a317:4adb:4b80::	2a02:a302:0:1::10	TCP	86	65421 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
1528	63.992318700	2a02:a317:4adb:4b80::	2a02:a302:0:1::10	TCP	86	64684 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
1529	63.992425000	2a02:a317:4adb:4b80::	2a02:a302:0:1::10	TCP	86	53397 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
1530	63.992498700	2a02:a317:4adb:4b80::	2a02:a302:0:1::10	TCP	86	60677 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM